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Mean radiant temperature – A predictor of heat related mortality

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ABSTRACT

Health studies have repeatedly used air temperature (T_a), sometimes adjusted for humidity, when analyzing the impact of weather on mortality. The aim of this study is to highlight the importance of mean radiant temperature (T_{mrt}) and its impact on heat related mortality. T_{mrt} is an essential meteorological parameter that influences the thermal comfort (heat load) of humans. It is useful when assessing the impact of weather, especially heat, on people's health. T_{mrt} is directly influenced by urban geometry and surface material, which also makes it a good measure to identify urban hot spots. The performance of models using T_a and T_{mrt} for daily mortality is compared for Stockholm County, Sweden. It is demonstrated that T_{mrt} models fit heat related mortality better than T_a models, which implies that health studies should consider using T_{mrt} rather than T_a. The use of T_{mrt} models allows us to determine more accurate thresholds for increased risks of heat related mortality, and thus to better identify adverse weather conditions and heat prone urban geometries. Such information is needed to implement heat-warning systems and mitigate harmful effects of heat stress.

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1. Introduction

Intense heat waves cause severe and devastating mortality and morbidity, especially in urban areas, due to local climate conditions (Jendritzky and Grätz, 1999; Pascal et al., 2006; Fouillet et al., 2003; Dousset et al., 2011; Gabriel and Endlicher, 2011). The elderly are most affected by risk factors related to health conditions or social environment, such as pre-existing diseases or social isolation (Kovats and Hajat, 2008). For example, the 2003 heat wave in Europe caused an estimated 15,000 excess deaths in France, of which 70% were in the age range of 75–94, with the largest part in Paris (Pirard et al., 2005; Le Tertre et al., 2006). It is predicted that cities will experience an increase in year-to-year variability of summer temperatures over the next century as a result of anthropogenic climate change, leading to a significant augmentation in the occurrence, intensity and duration of heat waves, particularly at mid-latitudes (Schär et al., 2004; Meehl and Tebaldi, 2004). The potential effects of climate change, together with the ageing of the population and on-going urbanization are expected to exacerbate the threats posed by heat stress in the future (McMichael et al., 2006; Oudin Åström et al., 2013). As a result, it is important to improve our understanding of the impact of weather on health and identify adaptation measures for societies to better cope with heat stress.

Heat exposure can generate heat stress as the human body attempts to retain normal temperature through sweating, often resulting in electrolyte imbalance, dehydration, and sometimes in heat stroke (Parsons, 2003). A decrease in temperature at night-time, which corresponds to lower physiological stress, allows for recovery. However, long episodes of heat exposure that exacerbate pre-existing cardiovascular and respiratory diseases, contribute to heat exhaustion, possibly leading to death, especially among the elderly (Kovats and Hajat, 2008). The time lag between exposure, stress, and death may range anywhere from a few hours to a few days (Kovats and Hajat, 2008; Rocklöv and Forsberg, 2008; Baccini et al., 2008). Temperature related mortality has increased considerably compared to "normal mortality levels" even on days that are not considered "heat waves" (Hajat, 2006; Rocklöv et al., 2011; Anderson and Bell, 2011). A few studies have differentiated the effects of heat intensity from those of heat wave duration and demonstrated contrasting impacts on population groups (Rocklöv et al., 2012; Barnett et al., 2012). Besides the elderly and people living in social isolation, individuals afflicted by pre-existing diseases or taking specific drugs or medications that affect perception or regulation of heat in the body are especially vulnerable to heat stress (Kovats and Hajat, 2008; Åström et al., 2011; Bouchama et al., 2007; Hajat et al., 2010).

One of the most important meteorological parameters governing the human energy balance and thermal comfort (heat load) is the mean radiant temperature, T_{mrt} (Mayer and Höppe, 1987; Ali-Toudert and Mayer, 2007). The T_{mrt} is defined as the 'uniform temperature of an imaginary enclosure in which the radiant heat transfer from the human body equals the radiant heat transfer in the actual non-uniform enclosure' (ASHRAE, 2001). The radiant (short-wave and long-wave) heat exchange between a person and their surroundings can be monitored or modelled. The most accurate way to monitor T_{mrt} is by measurements of short-wave and long-wave radiation flux densities from the three-dimensional surroundings of humans (east, west, north and south, upward and downward) and the calculation of angular factors (i.e. proportion of radiation received by the human body in each direction) (Höppe, 1992; Thorsson et al., 2007). If the radiation flux densities and angular factors are known, the T_{mrt} (°C) can be calculated using the Stefan–Boltzmann law:

$$T_{mrt} = \sqrt[4]{(S_{str}/(\varepsilon_p \sigma))} - 273.15 \tag{1}$$

with

$$S_{str} = \alpha_k \sum_{i=1}^{6} K_i F_i + \alpha_l \sum_{i=1}^{6} L_i F_i \tag{2}$$

where: S_{str} (Wm^{-2}) is the total radiant flux density absorbed by body; K_i and L_i are, respectively the short-wave and long-wave radiation fluxes densities in the six directions (i = 6); F_i is the angular factor (i = 6); α_k and α_l are, respectively the absorption coefficients for short-wave and long-wave radiation; ε_p is the emissivity of the human body and σ is the Stefan–Boltzmann constant.

Another method to monitor the T_{mrt} is by using a globe thermometer combined with measurements of air temperature and wind speed (Thorsson et al., 2007). The T_{mrt} can also be estimated by using models such as ENVI-met (Bruse and Fleer, 1998), RayMan (Matzarakis et al., 2009), or SOLWEIG (Lindberg et al., 2008; Lindberg and Grimmond, 2011). In this case, meteorological data from standard weather stations are used as input and the urban environment (buildings, trees, etc.) surrounding the site is modelled.

Whereas intra-urban T_a differences are rather small during the day, the T_{mrt} shows large spatial variations over short distances (Emmanuel and Fernando, 2007; Mayer et al., 2008; Lindberg and Grimmond, 2011; Thorsson et al., 2011). These variations are mainly governed by shadow patterns generated by obstructing objects such as trees, buildings and topography, but also by differences in thermal and radiative properties of the surrounding surface materials, i.e., albedo, emissivity, and heat capacity. During clear summer days with high irradiance, the highest T_{mrt} is found in areas near sunlit walls at noon/early afternoon, as a result of the high amount of direct and reflected short-wave radiation, and emitted long-wave radiation from the neighboring surfaces exposed to the sun. On these days and in these locations, T_{mrt} may be substantially higher than T_a. At night, when short-wave radiation is absent, T_{mrt} is more or less equal to the T_a.

On clear summer days, radiant heating is an important heat load source on humans whether indoor or outdoor. In the latter case, the load is greatest when humans occupy sunlit spaces. When indoor, part of the solar radiation absorbed at building surfaces is conducted through the walls and roof and transmitted through windows raising indoor air temperature. The response is a function of building design, the radiant and thermal properties of its materials, its insulation and external shading (Givoni, 1988; De Dear and Brager, 2001; Pollard et al., 2001).

Although, some recent attempts to evaluate the relationship between heat stress and mortality have been achieved using indices based on the heat balance equation (rational indices) Muthers et al., 2010; Nastos and Matzarakis, 2012, simple measures such as maximum, minimum, or mean air temperature, sometimes adjusted for humidity have been used in most epidemiological studies to date (Basu and Samet, 2002; Baccini et al., 2008; Anderson and Bell, 2009; Rocklöv and Forsberg, 2010). Most of the epidemiological studies have found little variation in mortality related to the indices used. Although variations according to age, season, and location were reported (Barnett et al., 2010), more research is needed on the specific factors underlying and exacerbating the effects of the interaction of the physical environment and weather with human physiology and its deadly consequences (Kovats and Hajat, 2008). Most analyses of mortality and weather conditions study aggregated disease (e.g., mortality counts) over time using a time series approach (Armstrong, 2006). The often large and crude aggregation levels hamper the analysis of spatial contrast from urban heat islands within the studies thus the focus has remained on time varying exposures that are being measured at meteorological stations.

Knowledge of the heat load and its spatial and temporal variation according to urban environmental factors is essential to better understand the drivers of heat-related stress and mortality to design preventive measures and promote well-being in cities. The availability of well-performing, user-friendly models such as ENVI-met, RayMan and SOLWEIG enable the assessment of human heat load and identification of urban geometries prone to heat stress at different times of the day and year.

The overall objectives of this paper are to highlight the advantages of using T_{mrt} to assess the impact of heat stress on heat related mortality, recognize weather conditions that increase the risk of heat related mortality, and identify urban geometries related to hot spots. The specific objectives are as follows.

(i) Compare the performance of time series regression models using daily maximum and minimum T_a and maximum and minimum T_{mrt} to explain daily mortality in the county of Stockholm, Sweden between 1990 and 2002.
(ii) Determine critical thresholds at which the risks of heat related mortality are significantly rising.
(iii) Identify weather conditions that cause increased risks.
(iv) Detect the built geometries prone to heat within a typical built-up area in Stockholm.

2. Case study area

Stockholm County is located on the Baltic Sea coast of Sweden (Fig. 1). Of the 21 administrative counties in Sweden it is the most populated with 95% of the population living in urban areas (Statistics Sweden, 2013). From 1990 to 2002, its population increased from 1.6 to 1.9 million, presently reaching approximately 2.1 million. The county extends over 6488 km^2 and 180 km from north to south. About half of the land area is forested (46%) while the other half is comprised of agricultural land (18%), human settlements (14%) and other uses (22%). The indented coastline has 60 km wide archipelago that imparts both coastal and continental characteristics to the areas. During winter, when the sea is covered by ice, the transfer of heat and water from the sea decreases, rendering the climate of the region more continental.

The monthly daily minimum, maximum and mean T_a and T_{mrt} for Stockholm between 1990 and 2002, as well as the highest and lowest recorded T_a and calculated T_{mrt} using SOLWEIG1D (for more information on the model and observation data see section 3) are shown in Fig. 2. July is typically the warmest month, with relatively high daytime temperatures (average daily maximum T_a and T_{mrt} of 22.5 °C and 47.2 °C, respectively) and cool night-time temperatures (average daily minimum of T_a and T_{mrt} 14.5 °C and 10.5 °C, respectively). The largest difference between T_a and T_{mrt} occurs in late

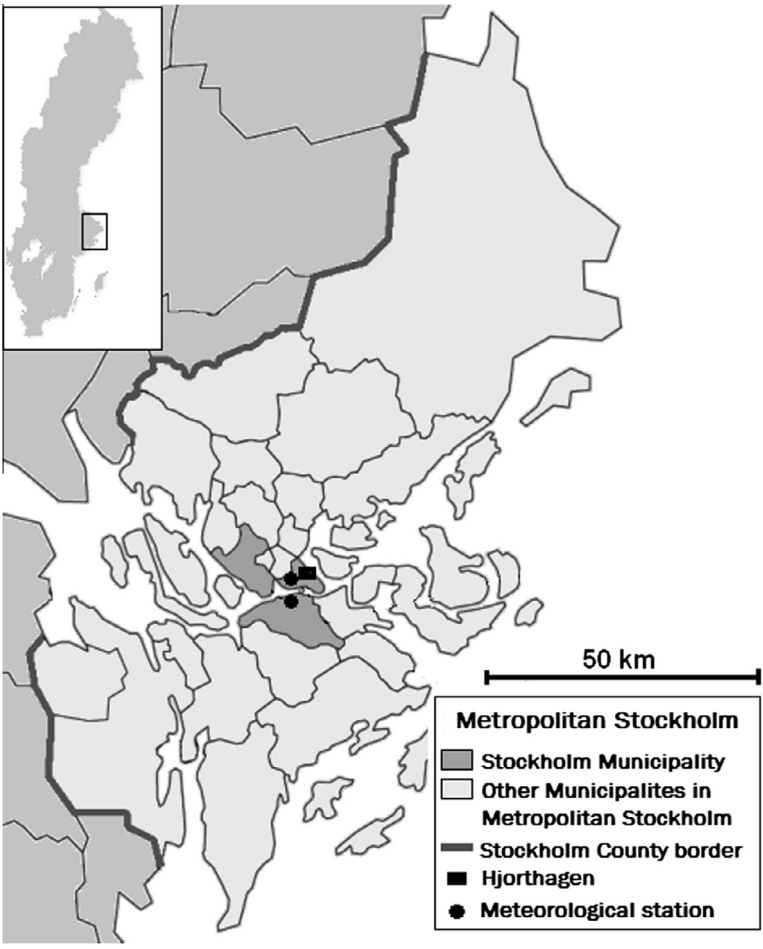

Fig. 1. Map showing Stockholm County, Hjorthagen and the location of the two meteorological stations.

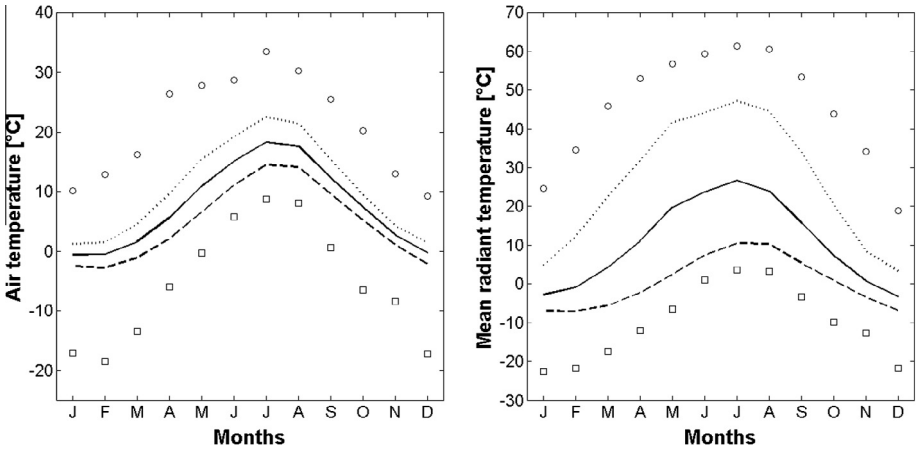

Fig. 2. Monthly course of average daily minimum, maximum and mean recorded air temperature (T_a) and calculated mean radiant temperature (T_{mrt}) for Stockholm between 1990 and 2002. The highest and lowest monthly T_a and T_{mrt} recorded during the period are shown as circles (○) and squares (□).

spring/summer on clear and warm days around noon. On these days differences of about 30 °C are common, but differences up to 38 °C have been recorded (not shown). The length of the nights varies greatly throughout the year, from approximately 6 h in June to 18 h in December.

A typical Stockholm built-up area (100 m × 200 m) was selected as a case study to analyze the influence of built geometry and vegetation on T_{mrt}. This area is situated in the Hjorthagen district, in the central part of Stockholm (Fig. 1). Hjorthagen is a former industrial and commercial harbor that is being redeveloped into a residential and commercial district. The building footprint and tree fraction are 0.23 and 0.13, respectively and mean building height is 18.6 m. Pervious, impervious and water surfaces are not accounted for.

3. Data and methodology

The data set includes daily mortality from Stockholm County and hourly meteorological data, from which the T_{mrt} was calculated using the SOLWEIG model. The data was processed and analyzed along the following steps.

The daily all-cause mortality data for three groups (all ages; 45–79 years; and 80 years and above) for Stockholm County (1990–2002) were obtained from the National Board of Health and Welfare. Slow variations and trends in mortality were filtered-out by subtracting a cubic spline function with 6 degrees of freedom per year (a total of 78 over 1990–2002). The data were also adjusted for calendar events such as national holidays and day of week. The remaining day-to-day mortality residuals were then modelled as a function of the mean of lag 0 and lag 1 daily maximum or minimum T_a or T_{mrt} using Poisson regression, allowing for over-dispersion. It has been shown that for the Stockholm area there is a delay between heat stress and subsequent mortality of 0–1 lag days (Rocklöv and Forsberg, 2008); similar short delays have been established internationally as well (Anderson and Bell, 2009). For a detailed description of lag associations and adjustment for time varying confounders in studies of heat and mortality see Barnett and Dobson (2000) and Armstrong (2006). The association between daily minimum and maximum T_a and T_{mrt} and mortality was modelled using a natural cubic spline with 6 degrees of freedom. Models were fitted and compared on the basis of the Akaike Information Criterion (AIC). The AIC is related to maximum likelihood measures, but penalizes more complex fits. Here we have used a negative AIC, therefore, the smaller the AIC value the better the model fits. When comparing alternative models, differences in AIC greater than 10 strongly indicates a better model fit, while a difference in AIC value of 3 or more indicates a substantial difference in the deviance explained by the models (Barnett and Dobson, 2000). Please note that the AIC values bear no relation to the ori-

ginal mortality values, however they are related to the number of parameters estimated, the over-dispersion parameters and the quasi-likelihood estimated to the models.

In order to assess the impact of weather on mortality on a county level, hourly meteorological data from 1990 to 2002 were selected for statistical analyses. These data were recorded at two meteorological stations in inner city environments in the centre of the county (Fig. 1). Data on short-wave radiation were obtained from a weather station run by the Swedish Meteorological and Hydrological Institute (SMHI). Global solar radiation was measured by a Kipp&Zonen CM11 pyranometer, and its direct component by an Eppley NIP pyrheliometer, replaced in 2000 by a Kipp&Zonen CH1. Data on air temperature and relative humidity were acquired from a weather station run by the Environment and Health Administration in Stockholm. Air temperature and humidity were measured by a Rotronic HC1 instrument placed in a naturally ventilated radiation shield mounted on the northern side of the building, 17 m above the ground and 3 m from the roof. Because of the station rooftop location the local/urban effect is assumed to be limited during the day, however there might be a local/urban effect during the night. The urban measurements used here are the best available indicator of the air temperature to which most people in Stockholm County are exposed, since 95% of the population live in urban areas. During the 1990–2002 period data were missing for 11% of days, particularly in July (16%). Missing values were not imputed.

The T_{mrt} was calculated using SOLWEIG1D (Lindberg, 2013), a subversion of the SOLWEIG model (Version 2.3) (Lindberg et al., 2008; Lindberg and Grimmond, 2011). SOLWEIG1D calculates the T_{mrt} for a person occupying a given location. The model requires inputs on geographical coordinates, weather observations, surface characteristics and body posture. Those variables include: latitude, longitude, and altitude; air temperature, air humidity, and solar radiation (global and diffuse); sky view factor (SVF), albedo, and emissivity; absorption coefficients for short-wave and long-wave radiation; and angular factors. In this study the T_{mrt} was calculated for a person standing/walking in an environment where the individual is constantly exposed to the sun, assuming a 0.6 general SVF. Angular factors for the standing/walking person were set to 0.22 for radiation fluxes from the four cardinal points (east, west, north and south) and to 0.06 for radiation fluxes from above and below. Standard values of absorption coefficients for short-wave and long-wave radiation were applied, i.e. 0.7 and 0.97, respectively. Albedo and emissivity for buildings and vegetation were set to 0.20 and 0.95, respectively, according to Oke (1987). There is no distinction between various land cover surfaces in the current version of the model. For detailed equations and model evaluation see Lindberg et al. (2008) and Lindberg and Grimmond (2011).

Thresholds of different risk increase levels (in 5% increments) were determined and used to identify weather conditions that caused increased risks for heat related mortality at the county level. They were also used to detect urban geometries prone to heat in a typical built-up area (i.e. Hjorthagen). The intra-urban spatial variations of T_{mrt} were analysed using the full SOLWEIG model (Version 2.3) (Lindberg et al., 2008; Lindberg and Grimmond, 2011). Transmissivity of short-wave and long-wave radiation through vegetation was set to 5% and 0%, respectively, according to Lindberg and Grimmond (2011).

4. Results

Here, we present the performance of the models using maximum and minimum T_a and T_{mrt} for the daily mortality of different age groups (all ages, 45–79, and 80+) in Stockholm County (1990–2002),

Table 1
AIC values for models using daily minimum and maximum air temperature (T_a) and mean radiant temperature (T_{mrt}) over lag 0–1.

Age group	Minimum		Maximum		Difference minimum T_a–T_{mrt}	Difference maximum T_a–T_{mrt}	Difference Min T_{mrt}–Max T_{mrt}
	T_a	T_{mrt}	T_a	T_{mrt}			
All ages	28592	27933	28582	27927	659	655	6
45–79	25058	24492	25056	24502	566	544	−10
80+	25673	25073	25673	25062	600	611	10

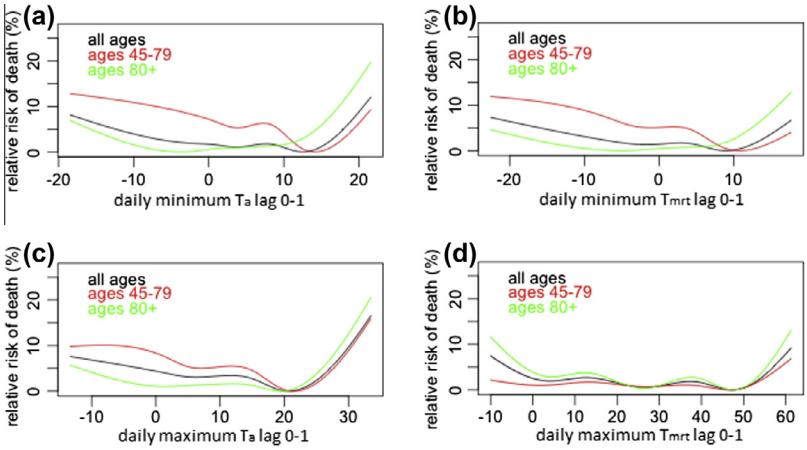

Fig. 3. Relative risk of death (%) and (a) daily minimum air temperature (T_a) (b) daily maximum air temperature (T_a), (c) daily minimum mean radiant temperature (T_{mrt}), and (d) daily maximum mean radiant temperature (T_{mrt}), lag 0–1.

and the thresholds for increased risks of heat related mortality. We also consider weather conditions that cause these increased risks and urban geometries prone to heat.

According to the AIC analysis, T_{mrt} is a much better predictor than T_a, for all age groups, since the estimated AIC difference is about 600 (Table 1). High daily maximum and minimum T_{mrt} are both associated with high relative risk of death (Fig. 3). For the 80+ group the daily maximum T_{mrt} is a better predictor compared to daily minimum T_{mrt} (AIC difference of 10, Table 1). For ages 45–79, however, the daily minimum T_{mrt} is a better predictor (AIC difference of −10, Table 1). This suggests that for the oldest segment of the population, heat related mortality is more attributable to daytime rather than nighttime heat stress, but for ages 45–79 heat related mortality is more attributable to nighttime heat stress. This also appears in the shape of the curves shown in Fig. 3.

The minimum mortality occurs when the daily maximum T_{mrt} ranges between 46 °C and 48 °C (Table 2 and Fig. 3d). Exposure to T_{mrt} above and below these values increases the relative risk of death. Mortality increases more steeply for high daily maximum T_{mrt} than for low daily maximum T_{mrt}. The eldest (80+) are most sensitive to heat. For this group, an increase in mortality is seen when daily maximum T_{mrt} passes above 47.6 °C. From 55.5 °C and onwards there is a significant risk increase (>5%). When the T_{mrt} passes above 59.4 °C the increased risk for heat related death is >10%. Note that it never exceeds 10% in the 45–79 group.

The highest T_{mrt} occurs on clear and warm days (Table 3). However, high T_{mrt} can occur at rather low T_a if combined with high global radiation and relatively low diffuse radiation, i.e., clear skies. For example, on a sunny day with a moderate T_a of about 20 °C, T_{mrt} can exceed 55.5 °C, thus causing a 5% increase in risk of death among the eldest group. As shown in Fig. 4, most hours (70%) with a $T_{mrt} \geqslant 55.5$ °C (i.e., when the increased risk exceeds 5% in ages 80+), occur between 1 and 2 p.m. Hours with $T_{mrt} \geqslant 55.5$ °C occur between May and August (Fig. 4) with most hours (66%) in July. This signifies that exposure to heat is greatest in summer early afternoon, when solar irradiance and emitted long-wave radiation is high.

Table 2
Risk classification, short-term effects of daily maximum mean radiant temperature (T_{mrt}) over lag 0–1.

Risk level	Risk increase (%)	Maximum T_{mrt} thresholds (°C)		
		All ages	Ages 45–79	Ages 80+
0	0	47.4	46.7	47.6
1	5	57.1	58.8	55.5
2	10			59.4

Table 3
Minimum and mean values of air temperature (T_a), global radiation (radG) and diffuse radiation (radD) with different risk increases and threshold values of daily maximum mean radiant temperature (T_{mrt}) for ages 80 + over lag 0–1.

Risk increase (%)	T_{mrt} (°C)	T_a (°C)		radG (Wm^{-2})		radD (Wm^{-2})	
		Min	Mean	Min	Mean	Min	Mean
⩾0	⩾47.6	9.9	21.6	185.1	641.3	32.6	132.4
⩾5	⩾55.5	20.5	26.4	465.2	675.7	48.8	119.1
⩾10	⩾59.4	27.2	29.0	564.6	688.5	63.4	99.3

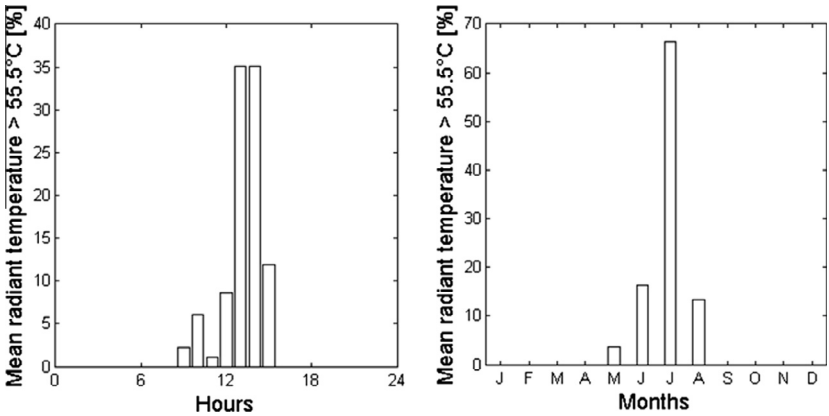

Fig. 4. Diurnal and monthly distribution of hours when mean radiant temperature (T_{mrt}) passes 55.5 °C (increased risk >5% in ages 80+).

Table 4 shows the average number of hours and number of days per year with different levels of risk increase for age groups 45–79 and 80+. Days with more than 5% risk increase occur on average 2 and 10 times per year, respectively, whereas days with more than 10% occur only once every year in the age group 80+. This means that days with moderate risk increase (⩾5%) among the eldest group occur relatively frequently, whereas days with more severe heat stress (risk increase ⩾10%) are comparatively rare.

Fig. 5 shows the spatial variations of T_{mrt} on a clear and warm summer day (10 July 1999) at 2 p.m. in the built-up area of Hjorthagen in Stockholm. Large spatial variations of T_{mrt} exist over short distances (up to 40 °C). Those variations are mainly due to the built geometry i.e. the dimensions and orientation of buildings and the intervening spaces and the trees, as these determine the amount of radiation reaching the surfaces (Fig. 5b). High T_{mrt} values occur in sunlit areas, particularly near the south and south-west facing walls. These high values also occur in narrow built structures, such as street canyons, due to the high amount of direct and reflected short-wave radiation, and emitted long-wave radiation from the neighboring surfaces exposed to the sun. Low T_{mrt} values are found in areas shaded by buildings or trees. Figs. 5c and d identify risk areas using the daily maximum T_{mrt} thresholds for ages 80 and older (Table 2) in the area without and with vegetation. A comparison of

Table 4
Average number of hours per year and number of days per year of different risk increases and threshold values of mean radiant temperature (T_{mrt}) for ages 45–79 and 80+ over lag 0–1.

Risk increase (%)	T_{mrt} (°C)	Ages 45–79		T_{mrt} (°C)	Ages 80+	
		Hours year^{-1}	Days year^{-1}		Hours year^{-1}	Days year^{-1}
⩾0	⩾46.7	324	70	⩾47.6	279	65
⩾5	⩾58.8	3.3	1.6	⩾55.5	23	10
⩾10				⩾59.4	2	1.2

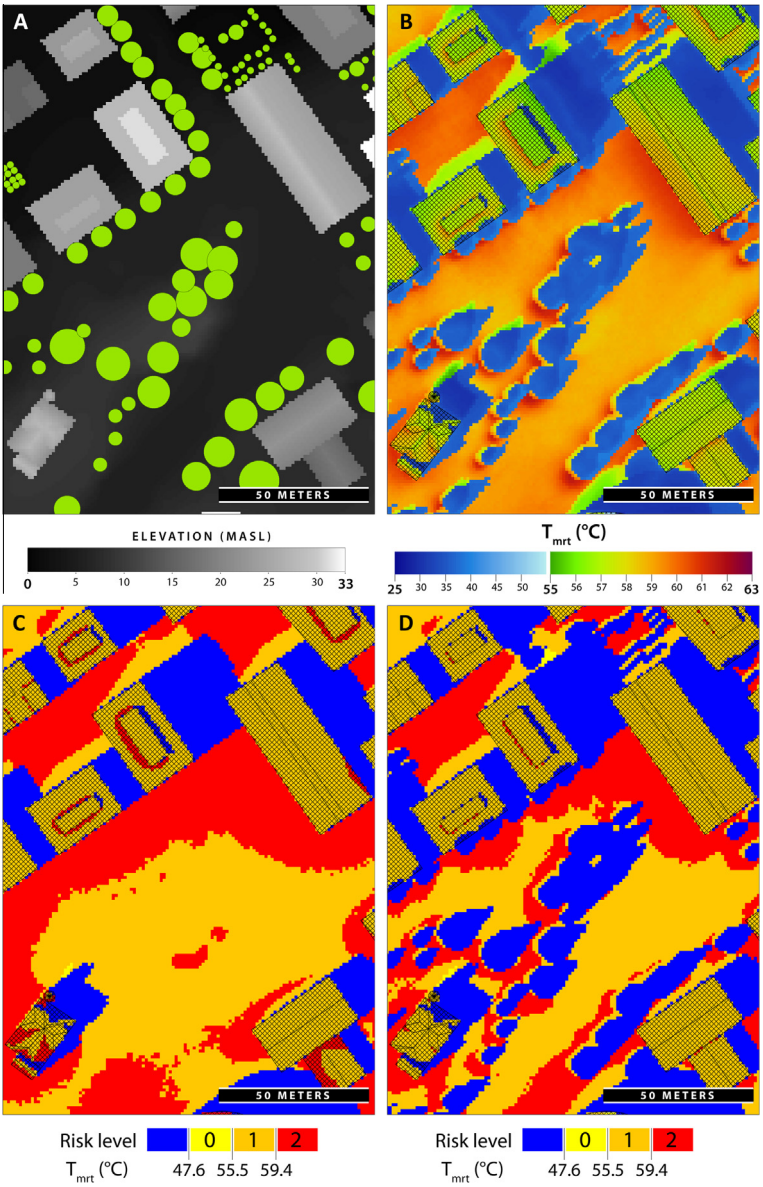

Fig. 5. Spatial variations of mean radiant temperature, T_{mrt} and identified risk areas in the built-up areas in Hjorthagen, Stockholm on a warm and sunny summer day (10 July 1999) at 2 p.m.; (a) digital surface model showing the elevation in meters above sea level (MASL); (b) T_{mrt}; (c) risk areas (without vegetation); and (d) risk areas (with vegetation). The green circles represent tree canopy foot prints. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the risk maps indicates that trees can be an effective measure for reducing outdoor heat stress in urban areas, by up to 30 °C. The shading effect provided by trees is about the same as that provided by buildings. However, at noon or in early afternoon the area shaded by the trees becomes rather limited due high elevation of the sun. Furthermore, in sun-lit locations the trees reduce the long-wave radiant

fluxes within their proximity, resulting in lower T_{mrt}, which is evident when comparing the extent of risk area 2 in Fig. 5c with that in Fig. 5d. This is due to the evapotranspiration at leaf surfaces which lowers the surface temperature of vegetation when compared to that of dry surfaces.

5. Discussion

The models using T_{mrt} provide a much better fit with heat related mortality than those using T_a (Table 1 and Fig. 3), which suggest that the T_{mrt} is a better predictor of heat related mortality than T_a. This result is attributable to the fact that T_{mrt} better describes the weather conditions that generates heat stress. Indeed, it includes not only the effect of air temperature (via emission of long-wave radiation from the sky and surrounding surfaces) but also that of solar radiation (direct, diffuse and reflected). Especially on clear summer days with high irradiance, T_{mrt} may be substantially higher than T_a, as result of absorption of solar and thermal radiation from the surroundings. For example, at noon/early afternoon the T_{mrt} can be 30–40 °C higher than the T_a in sun exposed outdoor areas. This implies that some heat stress situations, i.e. relatively low or modest T_a but with high irradiance may not be detected by simply looking at T_a. In other words it is inappropriate to use T_a as the main indicator for heat stress under hot and sunny conditions (Ali-Toudert and Mayer, 2007).

During clear days, radiant heating constitutes an important part of the heat load on buildings, causing an elevation of indoor air temperatures as the heat is conducted through the building (if not air-conditioned) (Givoni, 1988; De Dear and Brager, 2001; Pollard et al., 2001). In addition, as most Swedish buildings have natural draft or fan controlled ventilation, the "cool" indoor air is replaced by the "warm" outdoor air through the ventilation system, rising the indoor air temperatures. This means that people living in non-air-conditioned buildings might be exposed to heat stress both outdoors and indoors during periods of extensive heat. Elderly people who spend most of their time inside poorly insulated and/or ventilated homes (both day and night) are especially at risk.

The optimal temperature at which the mortality rate is minimal varies among the different age groups (Table 2). The eldest group (80+) is most sensitive to heat, i.e., their risk augments more with high T_{mrt} than for the other age groups. This result is in line with previous studies (Åström et al., 2011) and can be explained by the fact that heat tolerance decreases with aging when the physiological capacity for thermoregulation is reduced. Consequently, it takes a longer time for elderly people to recuperate from heat exposure and they are more likely to suffer from exhaustion. Their health may also be worsened by antecedent conditions such as cardiovascular and respiratory diseases, reduced mobility, mental illness, and the use of medication (Bouchama et al., 2007; Hajat et al., 2010). As shown in Table 4, days with a 5% risk increase among the eldest group occur relatively frequently (i.e., an average of 10 days per year), whereas days with more severe heat stress (risk increase $\geqslant$10%) are comparatively rare (i.e., an average of one day per year). The average daily mortality in Stockholm County during June to August for this age group is 19 deaths per day. This means that during an average summer there will be 10 excess deaths in this segment of the population as a result of heat exposure. Corresponding values for the age group 45–79 are two days and two excess deaths.

High daytime and nighttime T_{mrt} are both associated with increased relative risk of death (Table 1 and Fig. 3). However, for the 80+ population this risk is greater with high daytime T_{mrt} than with high nighttime T_{mrt}. In contrast, the risk for the 45–79 population is greater with high nighttime T_{mrt}. This suggests that during 'normal' summer weather, with high daytime temperatures and relatively cool nighttime temperatures, heat related mortality among the eldest is primary attributable to daytime heat stress, whereas heat related mortality among ages 45–79 is primarily attributable to nighttime heat stress. The different responsiveness of the age groups to high daytime versus high nighttime temperatures might be explained by their relative vulnerability to heat stress. For the oldest people (aged 80+) death by heat stroke or exhaustion may occur after a single warm day (Rocklöv et al., 2011). People aged 45–79 are more likely to die of exhaustion during long periods of consecutive hot days and nights, such as heat waves, that prevent them from recuperating at night from the heat stress experienced during the day (Rocklöv et al., 2011). Similar results were observed during the August 2003 heat wave in the Paris region. As in Stockholm, the risk of heat stroke attributable to high daytime temperatures, particularly at the start of the heat wave, was highest for people 80+ (Ravault et al., 2004). The mortality risk for people 65+, which represented 91% of the excess mortality, was closely

associated with the minimum temperatures (night), over the entire duration of the heat wave or over one week before death, whereas it was not significantly associated with the maximum temperatures (day) (Dousset et al., 2011; Laaidi et al., 2012). It was also shown that even a slight 0.5 °C increase during six consecutive nights doubled the risk of death (Laaidi et al., 2012). This 2003 extreme heat-wave caused an estimated 4867 heat-related deaths in the Paris region, mostly among elderly people. The highest mortality ratios corresponded to the districts with both higher nighttime temperatures and more socio-economical challenges. In those districts, the isolated elderly, who often live in poorly insulated buildings, did not have either the appropriate information or assistance to adapt their behavior or seek refuge in air-conditioned buildings (Fouillet et al., 2006; Dousset et al., 2011).

Our results from Stockholm County suggest that measures should be taken to reduce both daytime and night-time heat stress to prevent excess deaths among the elderly population (80+), who are most sensitive to heat. More studies are needed to better understand the daytime and night-time temperature contributions and interactions with the urban physical environment, and their human health effects, as function of both 'normal' summers and extreme heat events. For example, further studies are needed to identify the populations and population groups suffering from heat, the geographical determinants, and local risk factors contributing to excess heat related mortality in order to design and test preventive public health interventions.

During clear days the highest T_{mrt} is found in areas near sunlit walls (Fig. 5) at noon/early afternoon in July (Fig. 4). Therefore, measures to mitigate daytime outdoor heat stress should focus on reducing the radiant component in these areas, i.e., maximizing shading and reducing the absorption of heat into buildings/ground during these times of the day and year. Shading can be provided either by increasing the built density or by increasing the amount of urban greenery. Trees have been shown to efficiently block solar radiation, thus reducing the T_{mrt} (Ali-Toudert and Mayer, 2007; Lindberg and Grimmond, 2011; Konarska et al., 2013) as well as the surface temperatures of neighboring surfaces (Akbari et al., 1997; Shashua-Bar et al., 2009). The moisture availability from vegetation allows for a larger fraction of the net radiative flux to be balanced by evapotranspiration. For example, an increase of 1% in vegetation photosynthetic activity in cities such as London, Paris, or Los Angeles decreases the surface temperature of summer afternoons by 0.2 °C, 0.2 °C and 0.3 °C, respectively (Dousset, 1991; Dousset et al., 2011). During summer afternoons, the surface temperature in the parks of these cities are 2–5 °C cooler than the built surroundings depending on the size of park and the state of vegetation (Dousset and Gourmelon, 2003; Dousset et al., 2011). The cooling effect of trees is mostly due to shading, so vegetation should be placed near surfaces that tend to heat up, such as sun-exposed walls/roofs. This means that trees near north facing walls have a limited cooling effect on their surroundings. Adding vegetation in areas with no or little vegetation also produces a larger cooling effect than adding vegetation in already highly vegetated areas (Loridan and Grimmond, 2012). Another way to reduce the absorption of heat in cities is to increase the surface albedo (Rosenfeld et al., 1995). Such an increase has a major impact in reducing the heat absorption into buildings or ground, but a minor impact on daytime outdoor comfort conditions (Erell et al., 2014). Actually, the decrease in emitted long-wave radiation due to lower surface temperatures is more or less counterbalanced by the increase in reflected short-wave radiation at ground level. Night-time heat stress can be mitigated by maximizing nocturnal cooling, either by decreasing the built density or increasing the amount of vegetation and permeable surfaces, thus raising the radiative and evapotranspirative cooling (Holmer et al., 2007, 2012).

Rational human biometerological indices, such as the Physiological Equivalent Temperature (PET) and the Universal Thermal Climate Index (UTCI), which are used to assess the radiant, convective, and conductive heat load affecting the human body, might further improve the model fit as they also account for humidity, wind, and thermo-physiological parameters such as clothing, activity, age, etc. However, T_{mrt} has several advantages over these indices. For example, it requires less input data, which simplifies estimations. It is directly influenced by built geometry, surface materials, and vegetation, which makes it a good measure when estimating the efficiency of different strategies to reduce the radiant component. Although T_{mrt} is not yet well recognized by the general public and might be complex to interpret it is a very useful meteorological parameter. It allows for a better estimation of the thresholds for increased risks of heath related deaths, determination of weather conditions causing these increased risks, and identification of urban geometries prone to heat. Such information

is needed to implement heat-warning systems and prevent the harmful effects of hot weather and climatic changes in cities (Koppe et al., 2004).

The results from our work are specific to the city of Stockholm. However each city is unique in terms of latitude and climate zone, regional weather, geographical and cultural context, urban fabric, and physical characteristics. Furthermore populations differ in terms of demography, behavior, adaptation to a prevailing climate regime, and prevalence of risk factor diseases. For example, in warm cities people are more affected by cold temperatures, whereas in cold cities they are more affected by higher temperatures (Keatinge et al., 2000). Consequently, distinctive features should be considered when evaluating the impact of heat on people's health in different cities.

6. Conclusions

Models using T_{mrt} provide an overall much better fit with mortality than models using T_a, which implies that health studies should consider using T_{mrt} rather than T_a as they commonly do today. Compared to the T_a models, the T_{mrt} models allow researchers to more accurately describe mortality fluctuations, determine thresholds for increased risks of heat related deaths, and identify harmful weather conditions and urban geometries prone to heat. Given the capability of meteorological forecasting and the availability of well-performing, user-friendly models such as SOLWEIG to analyze the temporal and spatial variations of T_{mrt} in urban areas, such information should be applied to implement more accurate and timely health warning systems in cities.

During 'normal' summer weathers in Stockholm, with high daytime temperatures and relatively cool night-time temperatures daily maximum T_{mrt} provides a better model for heat related mortality than daily minimum T_{mrt} for the 80+ group. In contrast, high nighttime T_{mrt} is associated with higher risk in the age group 45–79. This suggests that heat related mortality among the elderly is primarily attributable to daytime heat stress, whereas heat related mortality among ages 45–79 is primarily attributable to nighttime heat stress.

In order to prevent the harmful effects of hot weather conditions on human health, measures to reduce the radiant component should be taken in areas prone to high T_{mrt}. High daytime T_{mrt} occurs mainly in the early afternoon on clear and warm days in narrow sunlit areas. As cool nights allow people to recuperate from the heat-stress experienced during the day, measures should also be taken to maximize nocturnal cooling. One of the few solutions that mitigate both daytime and nighttime heat stress is vegetation.

In future work, we will further investigate T_{mrt} measurements from different sensor systems and the complex exchanges between air, surface and radiant temperatures, as well as urban surface properties. We will expand the present work to different European cities, considering such effects as the health impacts of day versus night temperatures as a function of normal summer versus heat-wave event, and the temporal and spatial dynamics of risk exposure according to local climates and urban fabrics. We will also explore the relationship between heat related death and different human biometeorological indices, and the predictive performance of health related deaths according to distinct temperature measurements, in order to implement early warning systems.

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